Impact of hydrogen incorporation on performance and PBS instability in ultrathin ALD-InO_x FETs

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***Abstract*— We have investigated the impact of H-impurity in AlO_x-passivated ultrathin ALD-InO_x bottom-gate FETs on positive bias stress (PBS) behavior. While the devices fabricated on SiO_2 with N_2 annealing show better current characteristics and long-term stress stability, the devices fabricated on the ALD-deposited AlO_x dielectric show abnormal negative-V_{th} shift and change of stability behavior with N_2 annealing. These PBS instabilities in ultrathin InO_x FET are attributed to the H-incorporation from the gate dielectric.**

***Index Terms*- InO_x, ultrathin, hydrogen, AlO_x passivation, PBS.**

I. Introduction

InO_x-based oxide semiconductor (OS) is promising as a channel material for transistors embedded in back-end of line (BEOL) [1]–[3] thanks to the superior performance and on-off current ratio even fabricated at a low thermal budget. Recently, the OS channel formed by atomic layer deposition (ALD) gain much attention because of thickness controllability, high uniformity, and excellent step coverage for monolithic 3-dimension (M3D) integration [4]. Among OS materials for functional BEOL transistors, ultrathin InO_x channels realized by ALD system are considered to be a potential candidate attributing to its good electrostatics, tunable threshold voltage (V_{th}) and sufficient drive current even in thin body condition [5]. However, it is known that the trade-off between performance and instability is a critical issue in InO_x FETs [6], where the instability arose from the transistor operating atmosphere [7], intrinsic defects within OS channels [8] and impurity incorporation from the vicinity passivation layer [9]. Especially, hydrogen impurity diffused from the inter-layer dielectric and the high-k gate dielectric into OS channels leads to dramatic deterioration in electrical properties during device operation [10]. Prior studies have investigated the impact of hydrogen incorporation during bias stress in bottom-gate In-Ga-Zn-O (IGZO) FETs down to channel thickness (T_{ch}) of 2.6 nm, elucidating the physical mechanism of bias temperature instability (BTI) due to the hydrogen incorporation from the gate dielectric and highlighting an enhanced resilience against hydrogen effect in ultrathin OS channels [11]–[13]. On the other hand, these reports also experimentally proved the multiple roles of hydrogen for passivating the interface traps, which supported that a low concentration of hydrogen within OS channels can lead to improvement in subthreshold behavior [14], [15]. Although the passivation effect was highly emphasized in IGZO, the suitable concentration level of existing and incorporating hydrogen should be thoroughly analyzed in ultrathin InO_x FETs.

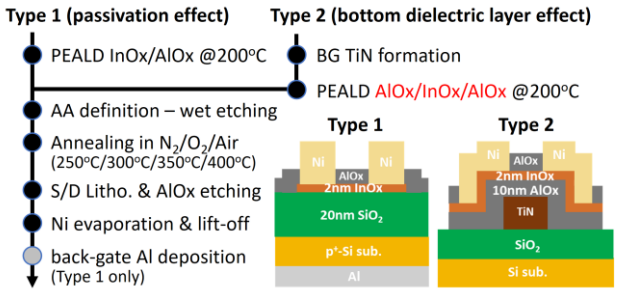


Fig. 1 Fabrication process flow and schematic diagram of AlO_x-passivated ultrathin InO_x FETs with heavily doped Si substrate (type-1) and TiN (type-2) for the bottom-gate structure, respectively.

In this study, we investigate the electrical and positive bias stress (PBS) characteristics of AlO_x-passivated ultrathin InO_x FETs annealed at different conditions. It is found that N_2 annealing is favorable for ultrathin InO_x channels. With a high quality of bottom SiO_2 oxide, hydrogens diffuse out of ultrathin InO_x channels during thermal annealing, which results in an improvement in device reliability. On the other hand, a high temperature annealing causes a pronounced hydrogen incorporation from the bottom AlO_x/TiN gate stack, which contains considerable amount of hydrogen, leading to the severe performance degradation and bias instability issues. Finally, a possible mechanism of H-induced instability is discussed.

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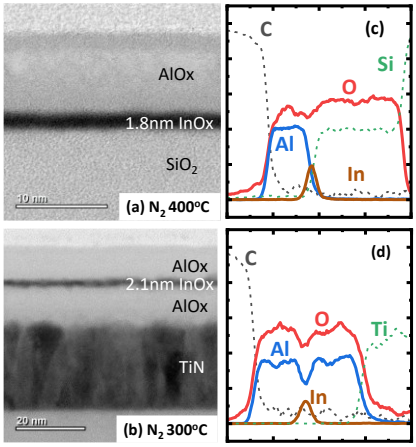


Fig. 2 TEM images of (a) AlO_x/InO_x channel annealed at 400 °C and (b) $AlO_x/InO_x/AlO_x$ channel annealed at 300 °C in N_2 ambient, and (c) and (d) corresponding EDX line-scan results of (a) and (b).

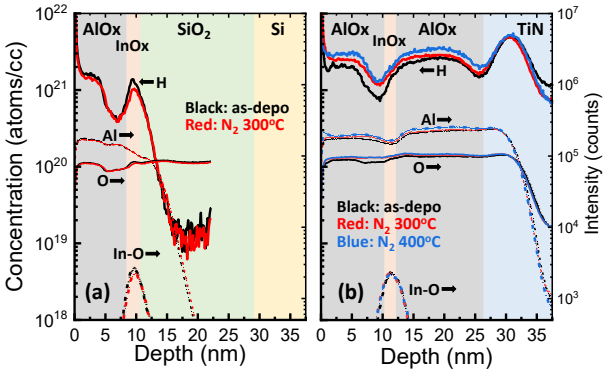


Fig. 3 SIMS analysis of (a) AlO_x/InO_x (type-1) and (b) $AlO_x/InO_x/AlO_x$ (type-2) structures with and without annealing in N_2 ambient.

II. Device Fabrication and Measurements

Fig. 1 shows the fabrication flow of AlO_x-passivated 2-nm-thick InO_x FETs with the heavily doped Si substrate (type-1) and TiN (type-2) for the bottom-gate structure, respectively. Firstly, a standard wet cleaning process was done on both type of substrates. Before the deposition of ultrathin InO_x film, 25-nm-thick TiN was deposited by sputter as the bottom-gate for type-2 structure, subsequently patterned by the ammonium wet etching process. The tri-layer of bottom dielectric AlO_x, ultrathin InO_x channel and surface passivation AlO_x layer were sequentially deposited by a plasma-enhanced ALD system with the fixed process temperature of 200 °C without breaking the vacuum. Noted that the deposition of the bottom AlO_x gate dielectric was skipped for the fabrication of type-1 devices. After film deposition, the active area was defined by HF-based chemical to simultaneously etch AlO_x and InO_x layers. Then, devices were split for the investigation on effect of annealing temperature (up to 400 °C) and ambient (N_2, O_2 or air atmosphere). For type-2 devices, we mainly focused on N_2 annealing in this study. The source and drain pads were then defined by selective wet etching of surface AlO_x and the deposition of 50-nm-thick Ni by lift-off process. Finally, 150-nm-thick Al was deposited at the back side of type-1 devices to form a better contact with heavily doped Si substrate. The dimension of all type-1 devices is the same with channel width of 25 µm and channel length of 35 µm. For type-2 devices, channel width and length of 25 µm and 10 µm with a gate to source/drain overlap area of 4 µm are designed, respectively.

To present the effect of electron trapping/de-trapping at the defect site, forward and backward current characterizations were performed with a fast-sweeping analyzer. The instability analysis was performed with a constant positive bias stress for all devices in this study [16].

III. Results and Discussion

A. Material analysis on AlO_x/InO_x channel after annealing

Fig. 2 show TEM images and EDX line-scan of AlO_x-passivated ultrathin InO_x FETs after 400 °C and 300 °C N_2 annealing in type-1 and type-2 structure, respectively. The estimated InO_x channel thickness is around 2-nm. It should be emphasized that the ultrathin InO_x channels remain amorphous even after 400 °C annealing, and there are no significant out-diffusion of In atom toward surface AlO_x passivation layer. To inspect the impact of hydrogen on electrical characteristics in ultrathin InO_x FETs, SIMS analysis focused on the hydrogen concentration of type-1 and type-2 structures before and after N_2 annealing are presented in Fig. 3. An out-diffusion of hydrogen from the ultrathin InO_x film is observed in the type-1 structure after 300 °C annealing in N_2 ambient, resulting in a reduction of total amount of hydrogen thanks to thermal treatment. On the other hand, hydrogen concentrations are increased after N_2 annealing at both 300 °C and 400 °C in type-2 structure due to the bottom gate stack, where hydrogen diffuses from the sputter-TiN into bottom AlO_x dielectric and incorporates into InO_x channel. Noted that a higher annealing temperature leads to a severer hydrogen incorporation into InO_x channel. It is expected that electrical characteristics and bias instability of AlO_x-passivated InO_x FETs will be significantly changed due to the out-diffusion or incorporation of hydrogen.

B. Electrical characterization on AlO_x/InO_x FETs

Fig. 4 shows first sweep I_d-V_g curves of type-1 devices with different annealing conditions. For N_2 and air annealing, FETs show increased maximum on-state current until the 300 °C annealing, which can be attributed to the increase of V_o and carrier after annealing in O-deficient ambient. However, electrical degradations are observed when further increase annealing temperature, which might be due to the severe degree of structural disorder in InO_x channel with too many V_o and deep-state traps [17], leading to the formation of acceptor-like site. These newly formed trap sites are considered to be the scattering center resulting in mobility and subthreshold slope (SS) degradation. Furthermore, a strong hydrogen out-diffusion at a high annealing temperature might lead to a weaker effectiveness of hydrogen passivation and a de-passivation of trapping sites within InO_x channels [15], resulting in a deteriorated SS characteristics as increasing annealing temperature higher than 300 °C. Therefore, the competition between free carrier generation and deep-state trap formation should be controlled by a medium temperature for N_2 annealing [18]. Devices annealed in O_2 ambient until 350 °C leads to deteriorated I_d-V_g characteristics, which might be due to out-

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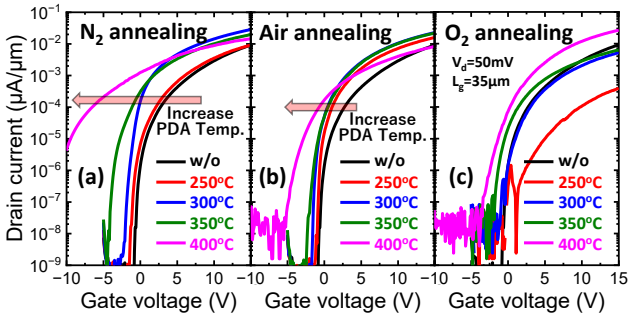


Fig. 4 Comparison of first sweep I_d-V_g characteristics of type-1 AlO_x-passivated InO_x FETs annealed by (a) N_2, (b) air, and (c) O_2 ambient with various temperatures. Noted that all devices are in the identical dimension.

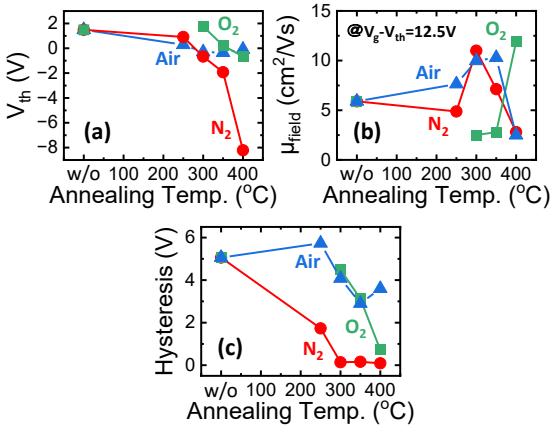


Fig. 5 (a) V_{th} values, (b) mobility results at overdrive of 12.5 V, and (c) hysteresis characteristics at V_{th} value in each type-1 device extracted from the first sweep of I_d-V_g curve.

diffusion of hydrogen and excess oxygen within InO_x channel [19], [20]. The excess oxygen will generate additional acceptor-like defect near valence band, where free electron at the conductive state might transmit to these low energy sub-band states resulting in electrical degradation. As the O_2 annealing temperature increases to 400 °C, an enhancement in drive current might be due to the transition of deep-state to shallow-state traps for hopping transportation without severe structural disorder suppressed by the oxygen. A further study is needed to investigate the effect of the high temperature O_2 annealing on electrical characteristics of ultrathin InO_x FETs.

Fig. 5 shows the extracted V_{th} values, mobility results and hysteresis characteristics of AlO_x-passivated InO_x FETs with type-1 structure after annealing in different conditions. The V_{th} value negatively shift as increasing N_2 annealing temperature, where a dramatic V_{th} shift occurs at 400 °C N_2 annealing due to severe structural disorder without enough hydrogen for defect passivation. Noted that devices annealed at 250 °C still show large hysteresis behavior, indicating that defects within ultrathin InO_x channel have not yet fully restored after a low temperature process. For air and O_2 annealing, negative V_{th} shift is suppressed thanks to incorporation of oxygen from ambient to suppress an increase of V_o, which shows a smaller V_{th} shift as compared to N_2 condition at an annealing temperature higher than 300 °C. However, comparing to the non-oxidizing N_2 annealing, oxygen incorporation into ultrathin InO_x channel causes additional defect generation for air and O_2 annealing at

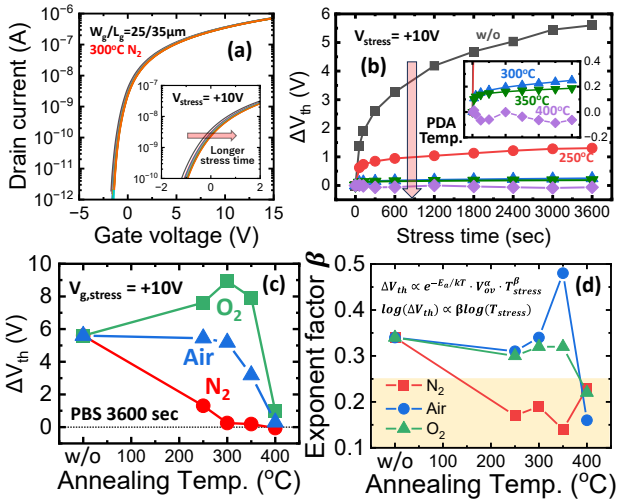


Fig. 6 (a) Evolution of I_d-V_g curves of 1-hour PBS test on AlO_x/InO_x FETs (type-1) annealed at 300 °C in N_2 ambient. The inset shows the enlarged curves near the V_{th} region. (b) The V_{th} shift trend of type-1 devices annealed in N_2 ambient with different temperature. The inset compares results of devices annealed at 300 °C, 350 °C and 400 °C. (c) The V_{th} shift value of type-1 devices after a constant PBS test for 1-hour, and (d) the degradation factor of β exponent parameter indicates the long-term PBS reliability.

a same temperature, making a drawback on device performance and hysteresis characteristics. These results suggest the issue of excess oxygen within ultrathin InO_x channels, which needs a higher annealing temperature to remove weakly bonded oxygen for improving electrical characteristics [21]. A large hysteresis behavior in devices with air and O_2 annealing indicates a poor bias instability due to acceptor-like defects generated from excess oxygen, which will not be feasible to device applications.

Fig. 6(a) shows the result of constant PBS test on AlO_x-passivated InO_x FET (type-1) annealed at 300 °C in N_2 ambient. It is found that V_{th} value rarely shift during PBS test and the excellent subthreshold characteristics is maintained after 1-hour stress. Fig. 6(b) compares the trend of V_{th} shift of type-1 devices annealed in N_2 ambient at different temperature during PBS test. The PBS instability is significantly improved after N_2 annealing at a temperature higher than 300 °C, which is consistent with the hysteresis characteristics shown in Fig. 5(c). Noted that devices with 400 °C N_2 annealing provide best PBS stability thanks to the reduction of hydrogen concentration, where the amount of hydroxyl bonds as a possible trapping site are reduced [15]. The negative V_{th} shift in device with 400 °C N_2 annealing might be due to the noise from measurement systems. Fig. 6(c) compares the 1-hour PBS instability of type-1 devices annealed in different conditions. As discussed, additional defect generated from excess oxygen leads to a severe instability issue in devices with air and O_2 annealing as compared to non-oxidizing N_2 annealing at a same temperature [19], [20]. A higher annealing temperature is needed to suppress the effect of excess oxygen which is compensated by the increase in amount of V_o from structural disorder of InO_x channels. Noted that the degree of excess oxygen issue in devices with air annealing is relatively minor as compared to O_2 annealing thanks to the lower oxygen partial pressure. Fig. 6(d) shows the stress time degradation factor of exponent β parameters [22], where devices with N_2 annealing show better long-term stability as compared to other one.

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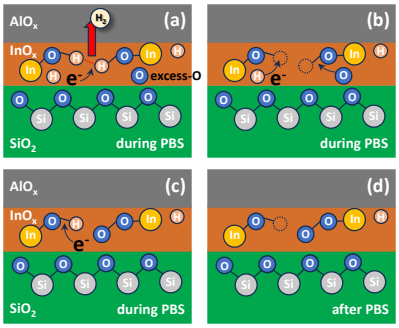


Fig. 7 Possible mechanism of PBS instability in type-1 devices without annealing, where (a) accumulated electron breaks two hydroxyl bonds leading to H_2 releasing, (b) migration of hydrogen interstitial and excess oxygen forming hydroxyl bond and O_2 dimer, respectively, (c) broken of newly formed unstable hydroxyl by accumulated electron, resulting in (d) vacancy and O_2 dimer as acceptor-like traps capturing free electrons.

Fig. 7 presents the possible mechanism of PBS instability in the type-1 device without annealing, containing higher hydrogen concentration within the InO_x channel in hydroxyl and interstitial sites. In the beginning, the existing hydrogen in In-OH bond was broken by the accumulated electron during PBS test, leading to the release of hydrogen as shown in Fig. 7(a). The existing excessive oxygen and hydrogen interstitial might migrate into vacancies sites (Fig. 7(b)) resulting in the acceptor-like O-O bond (O_2 dimer) and unstable hydroxyl bond. Furthermore, the newly formed hydroxyl bond might be broken by accumulated electrons again (Fig. 7(c)), leading to a vacancy defect as an additional trapping site within ultrathin InO_x channels, as shown in Fig. 7(d). As a result, the awful PBS instability of type-1 device without annealing attributes to the trapping site generation from the unstable hydroxyl bond because of the excess hydrogen, and the weakly bonded oxygen also devastates the reliability by forming O_2 dimer. Therefore, thermal treatments with a medium temperature will be necessary to reduce both hydroxyl bonds and hydrogen interstitials, suppressing the relating PBS instability in ultrathin InO_x FETs as proposed in Fig. 7.

C. Electrical characterization on $AlO_x/InO_x/AlO_x/TiN$ FETs

Fig.8 shows the forward and backward sweep of 1st and 5th I_d-V_g curves of $AlO_x/InO_x/AlO_x/TiN$ FETs (type-2) devices with and without annealing. The large hysteresis in 1st-sweep, resulting from trapping sites in gate dielectric AlO_x, can be reduced with increasing N_2 annealing temperature. However, the severe structural disorder of InO_x channels leads to the V_o generation and a negative initial V_{th} for 400 °C N_2 annealing, as discussed in the previous section. The on/off current ratio over 8-order is clearly observed in devices with pristine and 300 °C N_2 annealing conditions at V_d of 50 mV. On the other hand, devices annealed in O_2 ambient show deteriorated electrical characterization probably due to the newly generated defect from excess oxygen [19], [20]. Different from type-1 devices, effect of excess oxygen cannot be restored by O_2 annealing at 400 °C in type-2 devices, which might be due to the hydrogen incorporation from the bottom TiN with excess hydrogen interstitials as acceptor-like defect [15]. Further investigation will be needed to examine the correlation effect between excess oxygen and hydrogen in ultrathin InO_x FETs. Fig. 9(a) presents

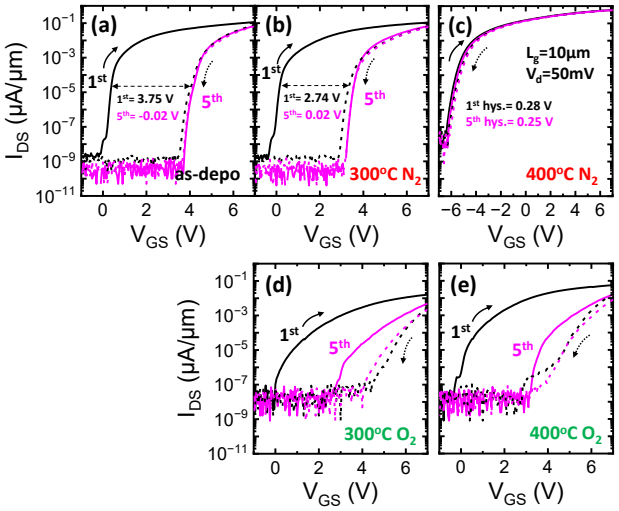


Fig. 8 1st- and 5th-sweep of forward and backward I_d-V_g of type-2 devices with (a) no annealing, (b) 300 °C N_2 annealing, (c) 400 °C N_2 annealing, (d) 300 °C O_2 annealing and (e) 400 °C O_2 annealing.

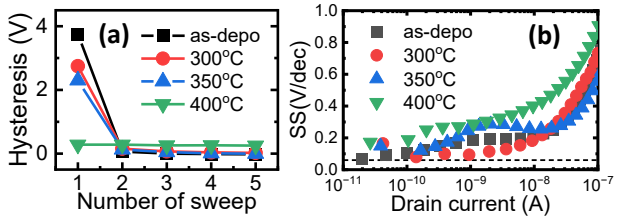


Fig. 9 (a) Hysteresis results of each I_d-V_g sweep and (b) SS value as a function of I_d of type-2 devices with different N_2 annealing temperature. The dash line in (b) indicates the ideal SS value of 60 mV/dec.

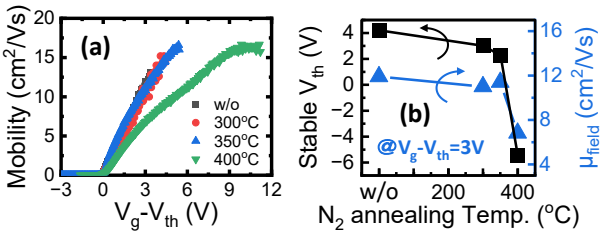


Fig. 10 (a) Mobility of type-2 devices with N_2 annealing as a function of overdrive voltage, and (b) the stable V_{th} and mobility results at overdrive of 3 V extracted from the 5th-sweep I_d-V_g curve.

the hysteresis voltage of type-2 devices after N_2 annealing at different temperatures. The hysteresis behavior become stable with <0.1 V after 3rd-sweep, indicating the saturation of trap sites within AlO_x after applying bias at a short time. It should be emphasized that devices with N_2 annealing at 400 °C show a larger hysteresis voltage even after few sweeping cycles, which might be due to a larger deep-trap states generated during high temperature process [17]. Fig. 9(b) shows the SS evolution of type-2 devices with different N_2 annealing temperature as a function of drain current. The minimum SS values of 67.7 and 79.5 mV/dec are achieved in devices with pristine and 300 °C annealing in N_2 ambient. As the hydrogen incorporating into ultrathin InO_x channels at a higher N_2 annealing temperature, the SS characteristic degrades due to the increase in amount of hydrogen interstitial following with acceptor-like trap sites. Fig. 10(a) shows the mobility as a function of overdrive voltage.

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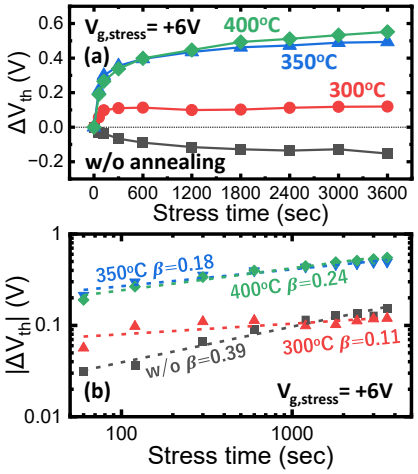


Fig. 11 (a) A constant PBS test of type-2 devices with and without N_2 annealing. (b) The value of V_{th} shift as a function of stress time for extracting exponent β parameters indicating long-term reliability.

There is no significant change on mobility trend after N_2 annealing until 350 °C. However, deep-trap state and acceptor-like defect generated from structural disorder and hydrogen incorporation at 400 °C annealing lead to the possible scattering center and mobility degradation. Fig. 10(b) plots the stable V_{th} values and mobility results at an overdrive voltage of 3 V extracted from the 5th-sweep I_d-V_g curve. Even though the negative V_{th} shift indicates an increase in carrier concentration of InO_x channels after 300 °C and 350 °C N_2 annealing, FETs provide a similar mobility value at certain overdrive. This implies the similar hopping transportation at a certain overdrive voltage among pristine, 300 °C and 350 °C annealing condition [23]. On the other hand, a dramatic change in the initial negative V_{th} and the strong scattering issue in FETs after 400 °C N_2 annealing reveals that AlO_x-passivated ultrathin InO_x channels are sensitive to annealing temperature.

Fig. 11 shows the result of PBS instability of type-2 devices with and without N_2 annealing. An abnormal V_{th} negative shift under PBS is observed in the device without annealing, which results from the hydrogen incorporation as well as doping effect during stress. Noted that the hydrogen concentration of bottom AlO_x dielectric is higher than InO_x channel, as shown by SIMS result in Fig. 3(b), indicating that the hydrogen keeps injecting from the beneath AlO_x as the positive bias is applied. Thus, the pristine devices show a poor long-term reliability of lager exponent β factor. On the other hand, the PBS instability behavior is changed after N_2 annealing, attributed to the existence of incorporated hydrogen within ultrathin InO_x channels. A higher annealing temperature results in a higher degree of incorporation, leading to a larger amount of hydrogen interstitial causing stability degradation. A V_{th} shift of 127 mV with excellent long-term reliability factor is achieved in AlO_x-passivated ultrathin InO_x FETs with N_2 annealing at 300 °C, showing the importance of process temperature and hydrogen incorporation from vicinity dielectric layer.

The possible mechanism of the abnormal PBS instability is presented in Fig. 12. Different from the SiO_2 case in the type-1 device, the defective bottom dielectric AlO_x contains a larger amount of hydrogen, which becomes a hydrogen supplier due

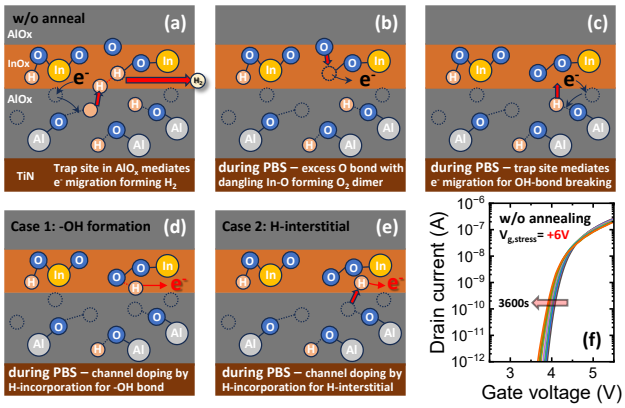


Fig. 12 (a)-(e) Possible mechanism of abnormal PBS instability and (f) I_d-V_g curves of the type-2 device without annealing under 1-hour PBS test.

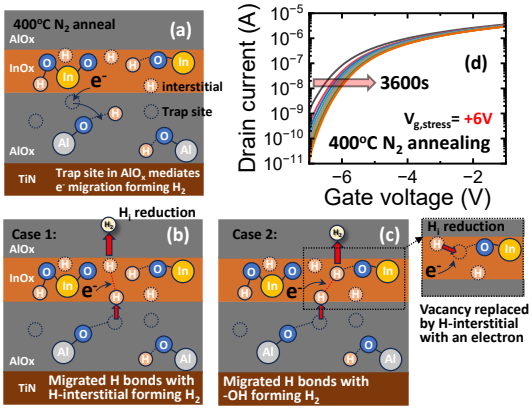


Fig. 13 (a)-(c) Possible mechanism of PBS instability and (f) I_d-V_g curves of the type-2 device with 400 °C N_2 annealing under 1-hour PBS test.

to electron trapping within AlO_x layers. During the PBS test, the accumulated electron will be trapped by the defect site at the AlO_x dielectric, which might further migrate and break the weak AlO-H bond leading to hydrogen releasing. The released hydrogen incorporates into the InO_x channel and bonds with hydroxyl site, resulting in the formation of out-diffusing H_2 gas, as shown in Fig. 12(a). It is considered that the pristine InO_x channel without annealing might contain excess oxygen, where the interstitial oxygen will replace the vacancy site forming the O_2 dimer and releasing additional electron [10], as depicted in Fig. 12(b). The additional electron (or accumulated electron) will again be mediated by trap sites within AlO_x dielectric and break another AlO-H bond with hydrogen releasing (Fig. 12(c)). Two scenarios might occur when the additional hydrogen incorporates into the InO_x channel: hydroxyl bond or interstitial hydrogen formation. In the first case, the incorporated hydrogen bond with O_2 dimer site forming hydroxyl bond as a dopant, releasing free electron as shown in Fig. 12(d). The other possible case (Fig. 12(e)) is that incorporated hydrogen might release an electron without bond formation, residing in the InO_x channel as the doping interstitial site. In both scenarios, the doping concentration of InO_x channel is increased and the V_{th} value is negatively shifted. Additionally, the increase of defect site in the AlO_x dielectric due to AlO-H debonding leads to the SS degradation, which can be observed from I_d-V_g curves of PBS test shown in Fig. 12(f).

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On the other hand, Fig. 13 shows the different mechanism of PBS instability in the type-2 devices with N_2 annealing, where a large amount of existing hydrogen interstitial incorporated from the beneath films modifying the instability behavior. Firstly, a similar phenomenon of AlO-H bond breaking occurred when PBS is applied, as shown in Fig. 13(a). The released hydrogen incorporating into the InO_x channel can either bond with interstitial hydrogen (Fig. 13(b)) or hydroxyl site (Fig.13(c)) forming the out-diffusion H_2 gas. In the case of bonding with hydroxyl site, the remaining vacancy will be filled by the nearby interstitial hydrogen with an electron. Therefore, the total amount of hydrogen within the InO_x channel is reduced, leading to the reduction of doping concentration in both cases. Furthermore, combining with the newly generated defect sites in AlO_x dielectric, a positive V_{th} shift is observed in annealed devices during PBS test, as shown in Fig. 13(d).

IV. Conclusions

We have investigated the hydrogen-induced PBS instability in AlO_x-passivated ultrathin InO_x FETs with T_{ch} around 2-nm. The amount of hydrogen within the InO_x channel and hydrogen migration from the vicinity dielectric layer during thermal process are two dominant factors causing the bias instability. An abnormal instability behavior is observed in pristine devices due to hydrogen incorporation from the AlO_x dielectric, which can be addressed by annealing in non-oxidizing N_2 ambient thanks to the removal of excess oxygen mediated by a medium amount of incorporated hydrogen. However, InO_x structural disorder and severe hydrogen incorporation become critical issues at a higher annealing temperature of 400 °C, resulting in a negative initial V_{th} and the mobility degradation. A precise control over thermal process will be important for ultrathin InO_x channels due to the susceptibility to hydrogen impurities.

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